

Dharma Sabapathy

Dharma.r.sabapathy@gmail.com | Austin, Texas

EDUCATION

The University of Texas at Austin, Cockrell School of Engineering | Austin, TX

B.S. Mechanical Engineering, Expected May 2030

Westlake High School | Austin, TX | May 2026

GPA: 3.86/4.00 | Top 10% | ACT: 35 | National Merit Scholar, Yale Book Award

Advanced Coursework: Multivariable Calculus, Linear Algebra, AP Physics C Mechanics and E&M, AP Chemistry, AP Computer Science A

ENGINEERING & RESEARCH EXPERIENCE

Robotics Intern | NASA ER4 at Johnson Space Center

June - August 2026

- Collaborated with engineers on the redesign of the front trunk assembly for NASA's Lunar Terrain Vehicle, resolving inherited CAD conflicts and refining the full sheet-metal exterior.
- Calculated gas-spring sizing and mounting geometry for automatic hood deployment; designed sheet-metal structure, billet mounts, gas-spring interfaces, and a latch/release mechanism in SOLIDWORKS.
- Participated in design reviews and delivered the completed CAD assembly and technical documentation for continued development and future fabrication.

Mechanical Engineering Intern | Elevate Robotics, Apptronik

June - August 2025

- Collaborated with multidisciplinary engineers on design reviews, subsystem integration, and troubleshooting.
- Led the design and development of a mechanical HMI subsystem from concept through CAD in Onshape, 3D printing, and final SLS prototyping; incorporated DFM considerations for injection molding at scale.
- Produced and presented technical documentation and design iterations to the engineering team.

Research Intern | NASA STEM Enhancement in Earth Science (SEES)

July 2025

- Researched commercial-scale quiet supersonic aircraft concepts and sonic-boom reduction using CAD and CFD as part of a national research team.
- Led aircraft configuration development and evaluated design geometry and sound reduction strategies with CFD.
- Presented research to NASA scientists and engineers and at the AGU 2025 Annual Meeting.

Team Captain & Design Lead | FRC Robotics 2468 Team Appreciate

2022-2026

- Led technical development for a 150-student robotics program coordinating mechanical design, subsystem integration, technical requirements, and competition readiness.
- Directed robot conceptualization, CAD, prototyping, design reviews, and assembly; coordinated mechanical projects across the design team and incorporated mentor feedback.
- Served as lead driver, collaborating with programming and mechanical teams to define, test, and troubleshoot subsystems.

LEADERSHIP & PROJECTS

Eagle Scout | Boy Scouts of America, Troop 70

2014-2026

- 3-time Assistant Senior Patrol Leader; Member of the Order of the Arrow

Chief of Staff | Westlake High School Student Council

2022-2026

- Led project planning, task delegation, and execution across the Senior Cabinet for school-wide initiatives.

FIRST LEGO League | 30842 Team Blue Jay Bots

2016-2022

- Developed a self-raising assistive walker that received a provisional patent and was selected as a Global Innovation Award finalist; presented the concept to Disney Parks engineers.

CEO and Co-Founder | NexTech Projects, 501(c)(3)

2023-2025

- Co-founded a technology nonprofit that refurbished and distributed over \$50,000 in donated electronics to seven organizations; built partnerships with corporate donors and nonprofit recipients and coordinated distribution.

TECHNICAL SKILLS

Mechanical Design: SOLIDWORKS (CSWA), Onshape | Sheet Metal, DFM, Assembly Design

Fabrication: FDM & SLS 3D Printing, Laser Cutting, Rapid Prototyping

Programming: Java